Exploiting Foundation Models for Costless Open-Vocabulary Semantic Segmentation of Optical Remote Sensing Images

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***Abstract*—Deep learning algorithms have driven substantial progress in remote sensing semantic segmentation. However, conventional approaches typically rely on predefined semantic categories, necessitating costly data annotation and model retraining when new classes are introduced. While large-scale vision-language models (VLMs), such as contrastive language image pre-training (CLIP), enable segmentation of arbitrary class with natural language guidance, their limited localization capability poses challenges for dense prediction tasks. This study investigates the potential of CLIP for semantic segmentation of optical remote sensing images (RSIs) and proposes LDSeg, a novel training-free framework that optimizes localization ability and visual–textual alignment through dual representation refinement. Through systematic analysis, we identify anomalous "singularity" tokens in the CLIP visual encoder that aggregate global contextual information while disproportionately attracting attention from local patch tokens, thereby degrading spatial discriminability. To address this, we introduce a singularity feature repair (SFR) strategy that mitigates feature distortion by recalibrating these tokens. Furthermore, we develop a hierarchical semantic expansion (HSE) method to generate precise hierarchical text descriptions, enhancing cross-modal alignment. The SFR and HSE strategies complement existing methods, providing further improvements. Extensive experiments demonstrate that LDSeg achieves state-of-the-art performance, delivering mean intersection over union (mIoU) improvements of 0.95%–7.84% over existing methods. The code is available at https://github.com/yeelinsen/LDSeg**

***Index Terms*—Contrastive language image pre-training (CLIP), optical remote sensing images (RSIs), semantic segmentation, singularities, training-free.**

I. Introduction

THE accurate perception and understanding of the content of optical remote sensing images (RSIs) is crucial to the development of Earth observations and geographic information systems (GISs). Semantic segmentation aims to identify and categorize each pixel within an image. This process provides a semantic-level perception of the information contained within RSIs, which is widely utilized in various applications, such as urban planning [1], intelligent agriculture [2], and land-cover mapping [3].

Received 15 August 2024; revised 27 January 2025, 3 April 2025, 12 December 2025, and 14 February 2026; accepted 18 March 2026. Date of publication 25 March 2026; date of current version 3 April 2026. This work was supported by the National Natural Science Foundation of China under Grant 62076033. *(Corresponding author: Fei Su.)*

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Digital Object Identifier 10.1109/TGRS.2026.3677380

Over the past several years, deep learning algorithms have revolutionized semantic segmentation in optical remote sensing imagery. Fully supervised methods [4], [5] investigated, with pretrained models on large-scale datasets typically leveraged for feature extraction of E_{img}. This process is illustrated in the top part of Fig. 1(a). To mitigate the heavy dependence on annotated data, extensive research efforts [6], [7], [8] have been devoted to reducing such reliance. Despite significant advancements, these methods encounter notable limitations in practical applications due to several challenges: 1) the high cost of pixel-level annotation; 2) the limited number of semantic categories; and 3) weak generalization capabilities. These limitations hamper the deployment of these models in a wide range of real-world scenarios, as visual concepts are not confined to the categories recognized during training but often incorporate new and unseen categories. Therefore, there is a pressing need for a more flexible and efficient semantic segmentation method capable of handling open-set categories.

Recently, a transformative influence has been exerted by vision-language models (VLMs) on computer vision and remote sensing research [9], [10], [11], [12]. Represented by contrastive language image pre-training (CLIP) [9], these VLMs are pretrained on a web-scale image–text pair corpus, where contrastive learning is leveraged to align images with their corresponding captions in the feature space. This pretraining paradigm provides the possibility for text-description-based segmentation of arbitrary categories, which is also known as open-vocabulary semantic segmentation (OVSS). The top part of Fig. 1(b) illustrates this process. However, such image-level pretrained models cannot be transferred well to the semantic segmentation task, as this task demands local fine-grained discriminability. Therefore, recent studies [13], [14], [15], [16] have explored how to fine-tune the visual encoder of VLMs via task-specific data, as shown in the bottom part of Fig. 1(b). Notably, when these types of methods [15], [16], [17], [18] are applied to new scenarios, a small amount of data generally must be annotated and parameter fine-tuning must be conducted.

In contrast, some studies have focused on a more challenging research direction; training-free approaches that eliminate the need for additional data, training costs, and degradation of open-vocabulary capabilities [19], [20], [21]. [21] modifies the last layer of the CLIP visual encoder by directly employing value vectors. Li et al. [22] and Bousselham et al. [23] introduce additional pathways that compute self-self attention in parallel with the visual encoder layers and use

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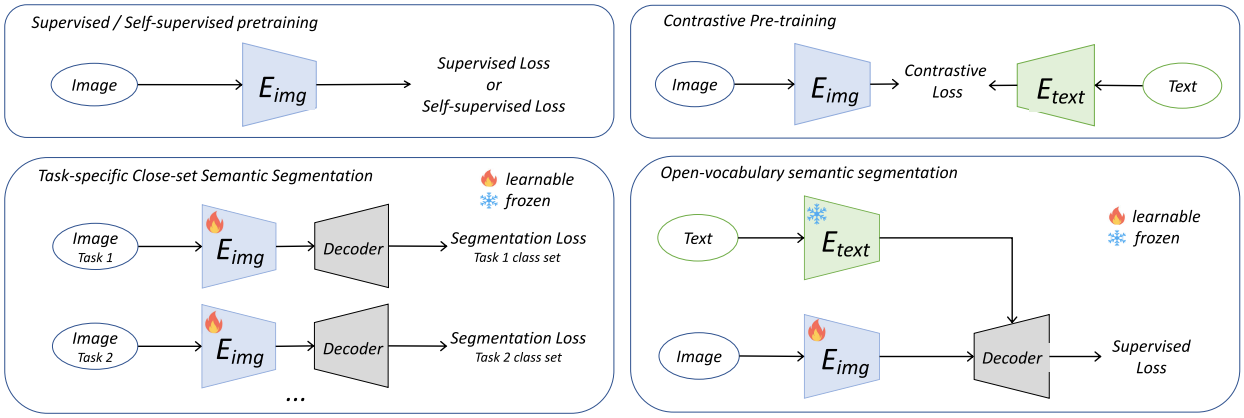


Fig. 1. Illustrations of different paradigms for semantic segmentation. (a) Closed-set paradigm methods, in which the top part indicates the pretraining of an image feature extraction model through a supervised or self-supervised approach; the bottom part represents the conventional closed-set methods. (b) Open-set paradigm methods, in which the top part is the contrastive language-image pretraining process and the bottom part depicts the adaptation of VLMs for the OVSS task.

residual connections to aggregate the outputs of multiple layers. Additionally, Li et al. [22] identified the presence of noisy activation in visual features, treating these noisy features as "redundant features" and performing "surgery" to mitigate their deleterious effects. These training-free methods focus mainly on modifying image encoders but overlook the significance of textual descriptions. This article demonstrates that appropriate textual prompts are crucial to language-driven semantic segmentation.

Despite advancements in the application of these methods to natural images, their utilization in the field of remote sensing remains largely unexplored. In this article, the potential of CLIP for semantic segmentation of optical RSIs is investigated, and a novel training-free framework termed LDSeg is proposed. With minimal modifications, this framework adapts the CLIP model for dense prediction tasks while preserving its robust generalization capabilities.

Specifically, abnormal tokens that are referred to as "singularities" are identified within the CLIP visual encoder through experimental analysis. These singularities encapsulate global information and disproportionately attract attention from normal patch tokens, which ultimately reduces spatial discriminability and degrades segmentation performance. The existence of such singularities is illustrated in Fig. 2. To address this issue, a singularity feature repair (SFR) strategy is proposed to rectify these anomalies. High-norm artifacts in supervised/self-supervised vision transformer (ViT) models [24] have been detected in a recent study [25], where this problem is resolved via the introduction of extra input tokens and retraining to smooth feature/attention maps. In contrast, our approach adapts CLIP for semantic segmentation without additional training or parameter adjustments. Furthermore, a straightforward hierarchical semantic expansion (HSE) strategy is proposed to generate more accurate textual descriptions for images, thereby improving the alignment between visual and textual representations. The main contributions of this article can be summarized as follows.

1) We introduce a novel, training-free framework, termed LDSeg, for the semantic segmentation of RSIs. The framework adapts the CLIP ability to align visual and textual features to dense prediction tasks while preserving its generalization ability.
2) We discover the inherent peculiar tokens within the CLIP visual encoder, characterized by both global information and high norms, which negatively impact semantic segmentation. We refer to these tokens as "singularities" and propose the SFR strategy to remedy the features of singularities.
3) We present a straightforward HSE strategy to produce more precise textual descriptions for images. This approach enhances the alignment between visual and textual representations, thereby improving segmentation performance.

This article is organized as follows. Section II thoroughly reviews the relevant literature. The key components of the proposed LDSeg are delineated in Section III. Performance evaluation and comparative analysis of LDSeg are presented in Section IV. Finally, Section VI summarizes this article and highlights the main conclusions.

II. Related Work

A. Closed-Set Remote Sensing Semantic Segmentation

In recent years, deep learning algorithms [4], [6], [26], [27], [28] have driven a revolutionary advancement in the semantic segmentation of optical RSIs.

Nevertheless, substantial human and material resources are demanded for data annotation, with pixel-level annotation being particularly labor-intensive and costly. To alleviate the reliance on labeled data while maintaining algorithm performance, research attention has been directed toward semisupervised learning approaches. Self-training [6], [29] is a popular methodology for addressing the scarcity of labeled data. Specifically, a model is first trained in a supervised

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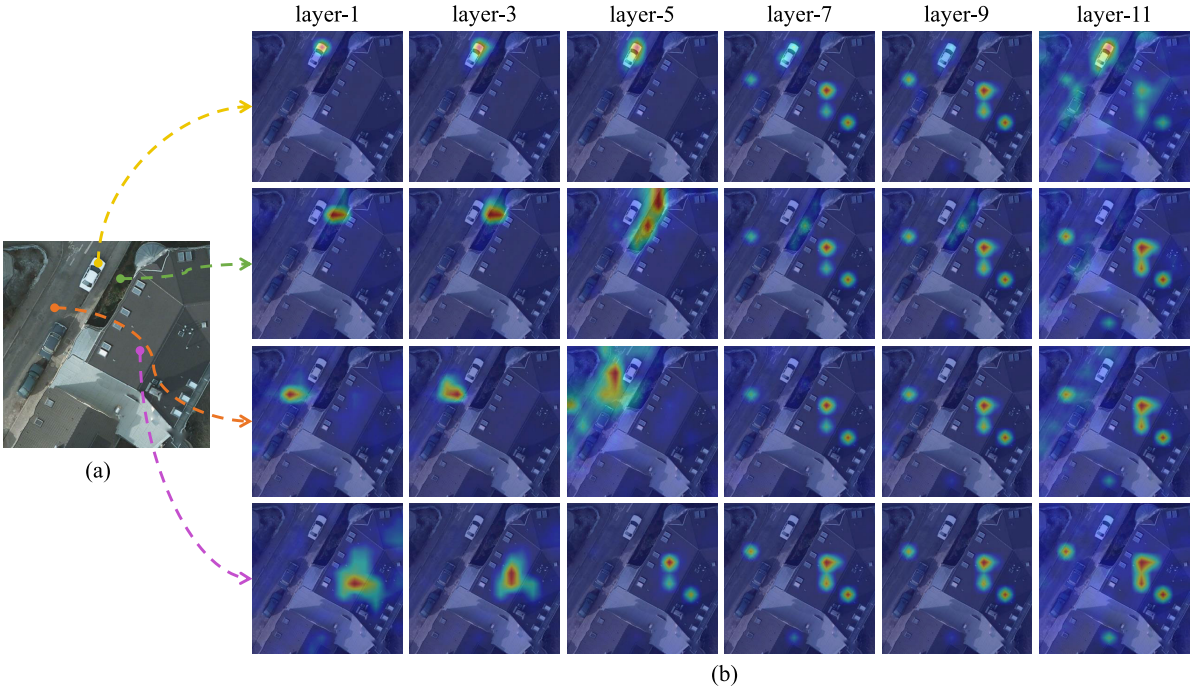


Fig. 2. Attention pattern evolution across network depths: shallow layers focus on local neighbors, whereas deeper layers attend to distant tokens, exhibiting homogeneous spatial distributions in late stages. (a) RSI, (b) illustrates the attention maps of different sampling points in different transformer layers.

manner, after which the trained model is utilized to generate pseudolabels for unlabeled data; iterative retraining is subsequently performed to enhance the accuracy. Meanwhile, consistency-based methods [28] leverage semantic invariance across data augmentations, apply consistency constraints to outputs from various augmentations, and often cooperate with techniques such as contrastive learning [30].

Recently, the segment anything model (SAM) [31] has attracted considerable attention within the remote sensing community owing to its impressive ability of segmenting images through diverse prompts. For example, RSPrompter [32] leverages a lightweight feature enhancer and two prompter structures, namely, anchor-based and query-based, to generate suitable prompts for SAM. Additionally, GeoSAM [33] performs fine-tuning on SAM using multimodal prompts, including point prompts from a pretrained model and text prompts from a large language model (LLM), along with Dice focal loss. Despite the notable progress achieved by these methods, their optimization tailored for closed-set datasets restricts their adaptability to real-world applications. This restriction stems from the fact that visual concepts transcend the predefined categories learned during training and often incorporate new and previously unobserved categories. When confronted with novel scenarios involving unseen categories, these methods generally require retraining or fine-tuning to achieve adaptation, a requirement that may impede their practical deployment in dynamic environments.

B. Language-Driven Semantic Segmentation

Language-driven semantic segmentation (LDSS) denotes a semantic segmentation paradigm that leverages textual information (e.g., descriptions and category names) to guide the segmentation process, enabling models to identify and segment target regions corresponding to arbitrary semantic categories as specified by language. Existing paradigms predominantly rely on vision-language model, typified by CLIP [9], harnessing cross-modal semantic correspondence between textual class descriptors and image feature representations for the localization of target semantic regions. Although CLIP [9] has exhibited impressive generalization capabilities in image-level tasks (e.g., classification and retrieval), it exhibits poor transferability to the semantic segmentation task, which requires local fine-grained discriminability. As a result, recent research efforts have focused on exploring strategies for fine-tuning the visual encoders of VLMs via task-specific data, as depicted in the bottom part of Fig. 1(b). OVRS [34] leverages a pretrained CLIP model for their open-vocabulary remote sensing segmentation method, which introduces a rotation-aggregative similarity computation module to handle orientation variations and integrates multiscale features into upsampling for scale-aware semantic masks. TPOV-Seg [35] proposes a textually enhanced prompt tuning approach, which includes a remote sensing-specific Text TempLator to enrich textual prompts, a lightweight text-aware prompt tuning strategy to align text encoders with remote sensing characteristics, and a textual-guided channel-aware aggregator (TGCA) to promote interchannel feature interaction. However, in practical applications where labeled data and ample training resources are unavailable, these methods often fail to produce satisfactory results. Moreover, models fine-tuned on specific datasets might lead to bias toward particular datasets, potentially forfeiting the open-vocabulary capabilities inherent in pretrained VLMs.

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C. Training-Free Approaches for LDSS

Some researchers explore more challenging training-free methods, which assert that they do not necessitate extra data or training expenses and do not diminish their capacity for open-vocabulary understanding [20], [21], [22], [23], [36], [37], [38]. While purportedly training-free, many of these approaches still rely on specific auxiliary tools, such as clustering models [36], mask proposal networks [20], and diffusion models [39]. Conversely, several studies [21], [22], [23], [37], [38] modify the architecture of the visual encoder while keeping the text encoder intact to maintain a broad vocabulary. Bousselham et al. [23] introduce additional pathways that compute self-attention in parallel with the original visual encoder layers and use residual connections to aggregate outputs from multiple layers. SCLIP [37] argues that the original self-attention module results in the misalignment of visual representations and proposes the correlative self-attention (CSA) mechanism. By substituting self-attention in the last transformer block with CSA, it enhances the semantic correlation and interpretability between visual features at different locations, significantly boosting the performance of semantic segmentation. NACLIP [38] suggests that previous efforts have neglected the significance of neighborhoods, thus introducing a locally enhanced neighborhood-aware model. This approach explicitly attends to the neighboring information of each patch by incorporating a Gaussian kernel to reinforce spatial consistency, further improving segmentation accuracy compared with that of [37]. To address challenges such as distorted target shapes and ill-fitting boundaries caused by low-resolution feature sensitivity, SegEarth-OV [40] introduces a feature upsampler (SimFeatUp) to restore the spatial information of deep features. These training-free methods focus primarily on modifying the image encoder, neglecting the significance of textual descriptions. In this work, we investigate the impact of textual descriptions and demonstrate that appropriate textual prompts can significantly improve segmentation performance.

III. Methodology

This section details the methodological design of LDSeg, a training-free framework for language-driven semantic segmentation of optical RSIs that introduces two key innovations to address CLIP's [9] limitations. Section III-A begins with a concise overview of CLIP to contextualize its adaptation for segmentation tasks. Following this, Section III-B presents the overall architecture and pipeline of LDSeg. Section III-C then reports the discovery of "singularities"—intrinsic anomalous features in CLIP's deep visual encoder layers. Subsequently, Section III-D analyzes the properties and impacts of singularities, alongside the proposal of the SFR strategy. Finally, Section III-E introduces the HSE strategy to refine textual descriptions and enhance visual–textual alignment.

A. Preliminary: Overview of Clip

CLIP [9] utilizes contrastive pretraining on web-scale image–text pair corpora to bridge vision and language. It comprises a visual encoder $\Phi_v(\cdot)$ and a text encoder $\Phi_t(\cdot)$, each processing their respective modality inputs. For text encoding, a text *prompt* is formulated via a prompt template such as "*a photograph of* $\{Text\}$," where $\{Text\}$ is the target category. Subsequently, the constructed *prompt* is encoded by $\Phi_t(\cdot)$, yielding the textual embedding $t = \Phi_t(prompt)$.

The visual encoding process, which adopts ViT [24] as the backbone, operates as follows. Initially, the input image $I \in \mathbb{R}^{H\times W\times 3}$ is divided into nonoverlapping $P \times P$ patches. Each patch is then vectorized into a local visual token $x_i \in \mathbb{R}^d$, where d is the feature dimension. Thereafter, these visual tokens are combined with positional embeddings and a [CLS] token that is responsible for capturing global image information to form a sequence $\mathrm{X} = \{x_{\mathrm{cls}}, x_1, \ldots, x_N\}$, where $N = (H/P) \times (W/P)$ denotes the number of visual tokens. Finally, sequence X is fed into L transformer blocks to output the visual feature sequence Z^{last}. The processing of input Z^l ($Z^0 = X$) by each transformer block can be formulated as follows:

$$\begin{aligned} q = Z^l W_q,\ \ k = Z^l W_k,\ \ v = Z^l W_v \\ SA = \mathrm{softmax}\left(q \cdot k^T / \sqrt{d}\right) \cdot v \\ Z^l = Z^l + \mathrm{SA}\left(Z^l\right) \\ Z^{l+1} = Z^l + \mathrm{FFN}\left(\mathrm{Norm}\left(Z^l\right)\right) \end{aligned} \tag{1}$$

where q, k, and v denote query, key, and value, respectively; W is the projection matrix; and Norm, FFN, and SA represent layer normalization, feed-forward network, and self-attention module, respectively.

When applying the CLIP [9] model to semantic segmentation, the task is reformulated as per-pixel/patch classification. Excluding the [CLS] token, the similarity between each visual token z_i (from Z) and each category-specific textual embedding t_j is computed, and the prediction for the ith patch is given by the following equation:

$$pred_i = \max_j \left(sim\left(\|z_i\|_2, \|t_j\|_2\right)\right) \tag{2}$$

where $\|\cdot\|_2$ denotes the L2 normalization and $sim(\cdot,\cdot)$ represents the cosine similarity.

However, CLIP's inherent design limits its ability to capture spatial localization information, resulting in unsatisfactory performance when directly applied to semantic segmentation tasks.

B. Training-Free Framework of LDSS: LDSeg

This section introduces the proposed LDSeg framework, a training-free solution for LDSS of optical RSIs. Built on the pretrained CLIP model [9], LDSeg adopts a ViT-Base [24] with L transformer layers as its visual encoder and a Transformer [41] as its text encoder, as illustrated in Fig. 3.

1) Correlative Self-Attention: As noted in [21] and [37], the main reason for the poor performance of vanilla CLIP [9] on the segmentation task is feature misalignment rather than insufficient feature extraction capabilities. In transformers, the position-related aspects are handled primarily in the self-attention module. To address this, improvements have been made to the self-attention module in CLIP [9]. Although each method has its own strategy, these improvements can generally

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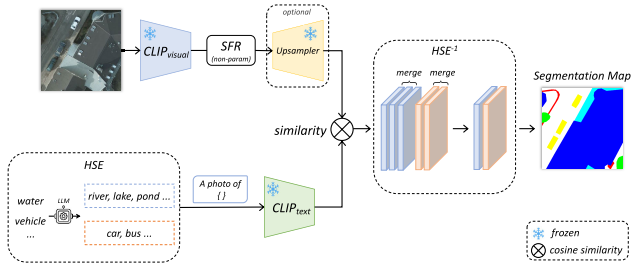


Fig. 3. Illustration of our proposed LDSeg, which comprises a dual-branch framework with a visual encoder and a text encoder. To optimize feature representation, the SFR method is integrated into the visual branch and an upsampler is optional to restore spatial details, while the HSE method is applied to the text branch for cross-modal alignment. (Zoomed-in view for better visualization of details.)

be seen as modifications that focus on the feature correlation between tokens that have passed through the same feature mapping function. We adopt the nomenclature from [37] and refer to these improvements as CSA. The main process of the CSA can be formalized as follows:

$$r = ZW_r \tag{3}$$

$$attn_{rr} = \text{softmax}\left(r \cdot r^T / \sqrt{d}\right) \tag{4}$$

$$\text{CSA} = \left(attn_{qq} + attn_{kk} + attn_{vv}\right) \cdot v \tag{5}$$

where W_r and $attn_{rr}$ represent the projection matrix and the self-self attention related to $r \in \{q, k, v\}$, respectively, and CSA is the correlative self-attention.

For visual embedding extraction, LDSeg slightly alters the last (Lth) transformer block of the CLIP visual encoder $\Phi_v(\cdot)$ while keeping the other parts unchanged; further details are subsequently presented. To be precise, given an input image I, the visual encoder first processes I up to the penultimate layer, yielding the intermediate feature sequence $Z^{(L-1)}$. This sequence is then fed into the modified last transformer block. The self-attention module is replaced with CSA to imbue visual tokens with positional and semantic information, and the feed-forward network is discarded in accordance with [22] and [23]. The operation of the last transformer block is formulated as follows:

$$\tilde{Z}^L = CSA\left(Norm\left(Z^{L-1}\right)\right) \tag{6}$$

where Z^{L-1} denotes the $(L-1)$th layer output.

Additionally, our experiments reveal intrinsic singularities in visual tokens, as depicted in Fig. 2. These singularities carry global information, attract excessive attention from normal patch tokens, and consequently reduce spatial discriminability and segmentation performance. To tackle this issue, we propose a feature repair strategy called SFR to rectify the visual encoder's output features, as detailed in Section III-D. The rectified visual features Z^L are then used for segmentation.

For the acquisition of text embeddings, LDSeg retains the original inference workflow of CLIP's text encoder. To fully leverage CLIP's vision-language capabilities, the HSE strategy is introduced to generate more precise text prompts, thereby enhancing image–text representation alignment. HSE is composed of two phases, symbolized by functions $\mathcal{H}$ and $\mathcal{H}^{-1}$. The first phase involves mapping superclasses to their corresponding subclasses by leveraging LLM, as indicated by $\mathcal{H}$; the second phase involves reverse mapping from subclasses to superclasses, as indicated by $\mathcal{H}^{-1}$.

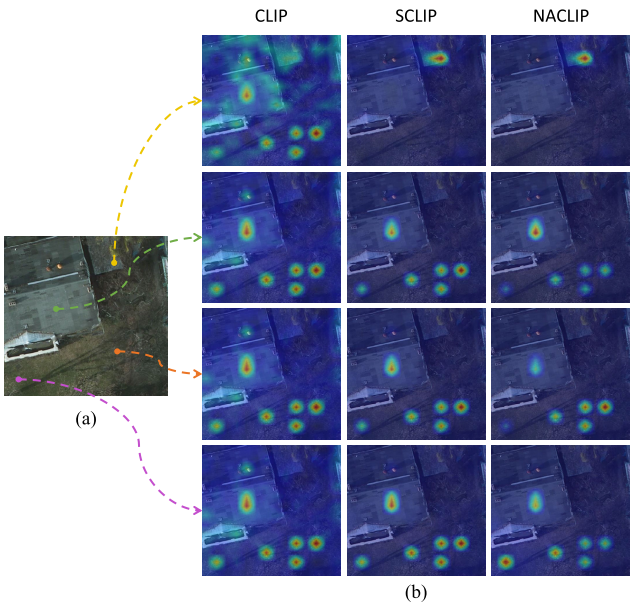


Fig. 4. Illustration of singularity. (a) RSI. (b) Attention maps of different sampling points in the last transformer layer. Each column from left to right in (b) attention maps of the last layers of CLIP [9], SCLIP [37], and NACLIP [38]. Each row in (b) attention maps of different sampling points, where the first row is the attention map of normal sampling points and the following three rows are the attention maps of the singularities.

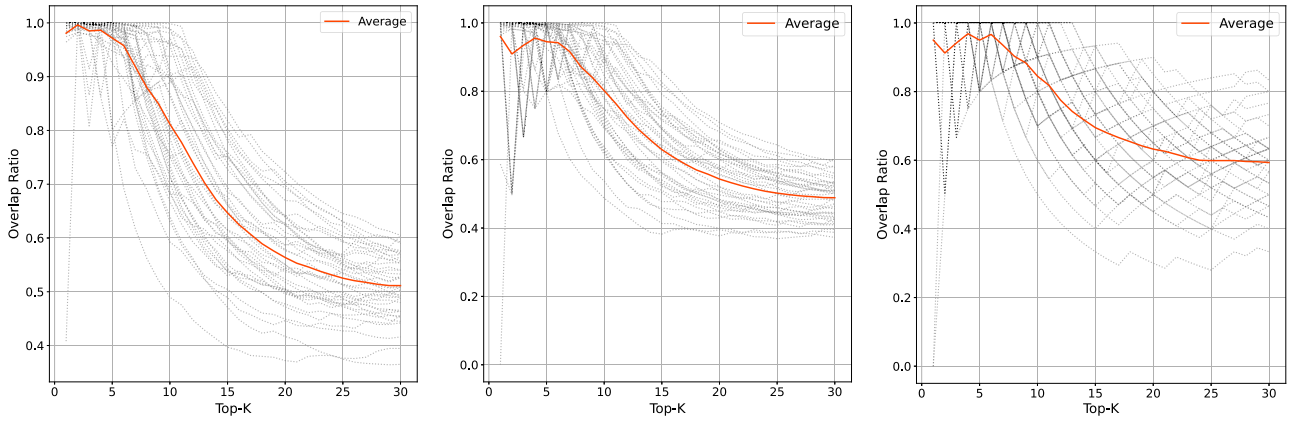
C. Singularity: The Devil in Visual Representations

In semantic segmentation, visual features are expected to preserve both discriminative capability and precise spatial structure. However, the original design of CLIP is not optimized for dense prediction tasks and tends to weaken positional sensitivity, which leads to degraded segmentation performance when directly applied. Previous studies [21], [37], [38] have shown that the standard self-attention formulation and training objective of CLIP contribute to reduced positional awareness in deeper visual representations.

To better understand the intrinsic properties of CLIP visual features, we conduct a set of diagnostic experiments to analyze the evolution of local representations across transformer layers. Specifically, we sample multiple spatial points from images across different datasets and examine their attention patterns. The investigation includes: 1) attention visualization across layers (Fig. 2); 2) comparison experiments with refined final-layer attention modules (Fig. 4); and 3) similarity and norm distribution analyses (Fig. 5). These experiments collectively evaluate how refined attention mechanisms influence local feature behavior.

As shown in Fig. 2, sampled points primarily attend to nearby tokens in shallower layers, indicating locally structured representations. In deeper layers, however, attention patterns become increasingly similar across tokens and extend toward distant regions. This observation is consistent with [37]. In addition, we observe that a subset of visual tokens exhibits

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(a) Global Information (b) High Norm (c) Global Information & High Norm

Fig. 5. Characteristics of singularity. We establish several distinct groups, each with unique properties. The nature of singularities is explored by analyzing the connections between these groups. We sort the cosine similarity between the [CLS] token and all the patch tokens to obtain the set $Sim_{cls\cdot}$, as well as the cosine similarity between each patch token and all the patch tokens (including itself), deriving $Sim_{i\cdot}$. Moreover, we rank the norms of all the patch tokens to obtain $Norm_i$. (a) Overlap between the top K patches of $Sim_{cls\cdot}$ and $Sim_{i\cdot}$. (b) Overlap rate between $Norm_i$ and $Sim_{i\cdot}$. (c) Overlap between $Norm_i$ and $Sim_{cls\cdot}$.

highly similar attention patterns and unusually high cosine similarity with most other tokens. Although recent works [37], [38] introduce refined attention mechanisms to enhance positional sensitivity in the final transformer layers, these abnormal tokens persist even after attention refinement.

We refer to these abnormal representations as *singularities*. Singularities typically occur in regions with repetitive textures or low information density and exhibit strong similarity concentration across tokens. It should be noted that tokens with high similarity values are treated as *indicators* of potential singularities rather than strict defining criteria. A singularity reflects representation dominance that is weakly tied to spatial semantics, which can be detrimental to dense prediction tasks requiring precise spatial discrimination. This observation suggests that dominant tokens emerge from deep-layer attention concentration and motivates the subsequent analysis of their structural properties.

Our observation is related to recent studies on attention concentration in multimodal models (e.g., Opera [42]). While both lines of work identify concentration phenomena in learned representations, their mechanisms and task contexts differ. Opera addresses hallucination in multimodal LLMs within generative settings and focuses on decoder-level token overtrust, whereas our work investigates encoder-side visual representation imbalance in vision-language segmentation models. The proposed analysis and correction operate on spatial feature representations rather than generative token prediction.

We further hypothesize that singularities may contain global information and/or possess large feature norms, which together lead to high cosine similarity with other tokens. To examine these properties, we conduct both qualitative and quantitative analyses.

1) Do Singularities Contain Global Information?: Since the [CLS] token encodes global context, patch tokens with high similarity to the [CLS] token are considered strong candidates for singularities. We compute $Sim_{cls\cdot}$ as the cosine similarity between the [CLS] token and all patch tokens, and $Sim_{i\cdot}$ as the similarity between each patch token and all patch tokens. We then evaluate the overlap between the top-K sets of these similarity rankings. As shown in Fig. 5(a), high overlap at small K values indicates strong global correlation of dominant tokens. Moreover, the overlap curves vary across images, and the inflection points—where overlap decreases rapidly—occur at different K values. This variability suggests that the number and prominence of singularities depend on image content.

2) Do Singularities Possess a Large Norm?: We compute $Norm_i$ by ranking patch tokens based on feature norms and evaluate its overlap with $Sim_{i\cdot}$. As illustrated in Fig. 5(b), dominant similarity tokens frequently coincide with high-norm tokens, indicating that large feature magnitude is another characteristic of singularities.

3) Are Singularities Characterized by Both Global Information and Large Norms?: Fig. 5(c) presents the overlap between $Norm_i$ and $Sim_{cls\cdot}$, demonstrating that many singularities simultaneously exhibit strong global correlation and large norms.

These findings indicate that the performance limitation arises from representation imbalance caused by singularities. Although singularities partially encode global information, their excessive dominance biases feature interactions toward a small subset of tokens. This imbalance motivates the design of a repair mechanism that rebalances feature interactions to improve spatial discrimination while preserving meaningful global semantic structure.

D. Singularity Feature Repair

Building on the above analysis, we introduce SFR as a mechanism to rebalance feature representations while preserving the semantic structure necessary for segmentation. As detailed in Section III-C, singularities inherently possess two detrimental traits, namely, global information dominance and high feature norm. The global information embedded in singularities compels normal tokens to excessively attend to

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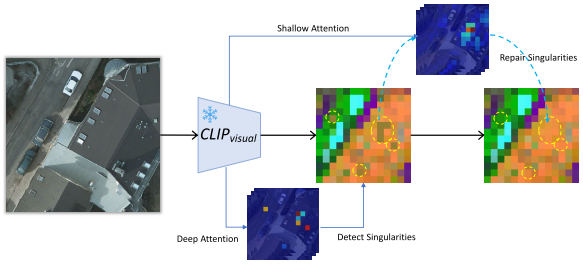


Fig. 6. Illustration of our proposed SFR method. (Zoomed-in view for better visualization.)

nonlocal, irrelevant regions, as visualized in Fig. 4. Such a phenomenon undermines the ability of normal tokens to focus on locally correlated areas, and spatial positional awareness is thereby eroded. Furthermore, although task-specific semantic information is lacking in singularities, the feature representation of normal tokens is still interfered with by them. Specifically, the feature distribution of surrounding normal tokens is distorted by their abnormally high-norm features, which results in ambiguous semantic discrimination. This ambiguity is evidenced by the overlapping feature clusters in Fig. 5. To this end, we propose a SFR method to identify singularities and then repair their features, as illustrated in Fig. 6.

1) Singularity Detection: Prior analysis in Section III-C reveals that singularities exhibit distinct distribution patterns compared to adjacent normal tokens. Abnormally high attention similarity is demonstrated by them with the [CLS] token, which encodes global image information, and among themselves. While conventional outlier detection algorithms such as kNN [43] and LOF [44] could identify such anomalies, additional computational overhead is introduced by these algorithms and tedious parameter tuning is required. Both of these aspects conflict with the efficient design goal of LDSeg. To balance detection accuracy and inference efficiency, a lightweight detection strategy is proposed, which leverages CLIP's internal attention signals and avoids dependencies on external modules. Concretely, the attention weights between the [CLS] token and all patch tokens are first extracted from the last three layers, with the final layer excluded. In this evaluation, the intersection of these similarities is considered until a noticeable difference is observed between any two layers

$$\begin{aligned} \mathrm{IDX}^i_{cls} &= \mathrm{argsort}\left(A^i_{cls\cdot}, descending = True\right) \\ \mathrm{SP} &= \mathrm{Intersection}\left(\mathrm{IDX}^i_{cls}\right) \end{aligned} \tag{7}$$

where $A^i_{cls\cdot}$ signifies the 0th row of the attention matrix of the ith layer, excluding the first element, which is the similarity between the [CLS] token and all patch tokens. IDX^i_{cls} denotes the indices of tokens ranked by their decreasing similarity to the [CLS] token of the ith layer, and SP is the set of singularities.

2) Singularity Repair: To endow singularities with accurate semantic and positional information, a feature repair strategy termed SFR is introduced. As established in Section III-C, singularities are predominantly localized in image regions with repetitive content and low information density, such as uniform farmland and continuous rooftops. For such regions, semantic consistency is typically shared by spatially adjacent tokens—an attribute that can be exploited to infuse meaningful local semantics into singularities. Notably, this repair logic is supported by the critical observation from Fig. 2; strong local correlation is exhibited by shallow-layer attention in CLIP's visual encoder, and normal tokens in shallow layers primarily attend to spatially neighboring regions rather than dispersing attention to distant areas, a phenomenon observed in deeper layers. This property renders shallow-layer attention an ideal source for mining semantically consistent features, which can be utilized to repair singularities.

Specifically, for each singularity $s \in \mathrm{SP}$, the attention weights are first extracted from the shallow transformer layers and then averaged. Layers 1–3 are selected as the shallow transformer layers based on the stability analysis and the resulting average is denoted as $A_{s\cdot}$. Thereafter, $A_{s\cdot}$ is sorted in descending order to identify the top N_f normal tokens that exhibit the highest attention similarity to the singularity; these tokens are deemed semantically consistent with the singularity's local context. Finally, the mean of the features of these top N_f tokens to replace the original feature of the singularity. The procedure of the SFR can be articulated as follows:

$$\begin{aligned} \mathrm{IDX}_s &= \mathrm{argsort}\left(A_{s\cdot}, descending = True\right) \\ \tilde{v}^l_s &= \frac{1}{\mathrm{IDX}_s} \sum_{k=1}^{N_f} z^l_{N_f[k]} \end{aligned} \tag{8}$$

where $s \in \mathrm{SP}$ is the index associated with singularity and $A_{s\cdot}$ signifies the sth row of the attention matrix, excluding the first element. IDX_s denotes the indices of tokens ranked by their decreasing similarity to the singularities, and $\tilde{z}^l_s$ is the synthesized feature in the lth transformer layer to displace the singularity feature.

E. Hierarchical Semantic Expansion

Earlier studies have dedicated efforts to optimizing image encoders for more effective visual representations, while the critical role of textual descriptions in LDSS has been overlooked. These studies typically employ simple text templates (e.g., "a photograph of {class}") paired with category names as textual prompts, which fails to capture fine-grained semantic nuances of remote sensing scenes. For instance, the superclass "building" aggregates residential buildings, factories, and sports facilities, which exhibit distinct spatial forms and scales in RS images; such vague descriptions lead to visual–text misalignment and degraded segmentation accuracy, as CLIP's pretraining lacks constraints to cluster subclasses of the same semantic group in the feature space.

To fully harness the text encoder's capacity and address this limitation, we propose an HSE strategy, which refines textual representations through two complementary phases: a forward mapping $\mathcal{H}$ from superclasses to subclasses and a reverse mapping $\mathcal{H}^{-1}$ from subclasses back to superclasses. This strategy generates precise, RS-specific textual prompts to

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enhance cross-modal alignment, while minimizing reliance on manual annotation.

Concretely, given a set of target superclasses $\mathcal{C}=\{c_j\}$, where each target category c_j is regarded as a superclass, an LLM is employed as the function $\mathcal{H}$ to automatically generate a set of subclasses $\mathcal{SC}_{c_j}=\mathcal{H}(c_j)$ for each target category c_j. To ensure consistency and quality of generated subclasses across datasets, we design a few-shot prompt template that incorporates the RS domain context and example outputs. This template guides GPT-3 [45] to produce subclasses, aliases, or visual attributes that are representative of RS scenarios, rather than generic natural image concepts. The prompt is formally defined as follows.

Task: For a given generic remote sensing category {class-name}, create related results (sorted by global usage frequency and remote sensing scene occurrence probability) with no cross-category ambiguity.

Q: List 3 common related results from multiple perspectives, including subcategories, aliases, and visual attributes for {airplane} in RSIs. *A:* Plane, aircraft, and jet planes. *Q:* List ten common related results from multiple perspectives, including subcategories, aliases, and visual attributes for {class-name} in RSIs.

After generating subclasses, we encode them into textual embeddings using CLIP's text encoder, following the same template as the original CLIP but with subclass names instead of superclass names. For each subclass $sc_j^n \in \mathcal{SC}_{c_j}$, we construct the text prompt as template(sc_j^n) and feed it into CLIP's text encoder $\Phi_t(\cdot)$. The resulting subclass text embeddings are denoted as follows:

$$\tilde{T}^{SC}=\Phi_t\left(\left\{\text{template}\left(sc_j^n\right)\mid sc_j^n\in\bigcup_{c_j\in\mathcal{C}}\mathcal{SC}_{c_j}\right\}\right). \tag{9}$$

With visual tokens X^L and text embeddings $\tilde{T}^{sc}$, semantic segmentation is approached as a per-pixel/patch assignment task. However, direct computation of the similarity between each visual token $x_i^L\in X^L$ and each subclass text embedding $t_j^n\in\tilde{T}^{sc}$ does not yield the final segmentation results directly. To project subclass logits back to the original superclass space, we introduce the inverse mapping $\mathcal{H}^{-1}$. Specifically, we average the logits of all subclasses belonging to the same superclass and then select the superclass with the highest averaged logit as the final prediction for z_i^L. This process is formalized as follows:

$$logit_{i,j}^n = sim\left(\|z_i^L\|_2,\|t_j^n\|_2\right) \tag{10}$$

$$logit_{i,j} = \mathcal{H}^{-1}\left(logit_{i,j}^n\right) = \frac{1}{|\mathcal{SC}_{c_j}|}\sum_{sc_j^n\in\mathcal{SC}_{c_j}} logit_{i,j}^n \tag{11}$$

$$pred_i = \arg\max_{c_j\in\mathcal{C}}\left(logit_{i,j}\right) \tag{12}$$

where $\|\cdot\|_2$ denotes L2 normalization, $logit_{ij}^n$ indicates the logit of the ith visual representation for the nth subclass of the jth superclass, and $pred_i$ is the final result in the original target classes $\mathcal{C}$.

IV. Experimental Results and Analyses

A. Datasets

Our proposed LDSeg is evaluated across seven land-cover mapping datasets, which consist of both satellite and high-resolution aerial imagery. The satellite image datasets include DLRSD [46], LoveDA [47], and iSAID [48], whereas the aerial image datasets include Potsdam, Vaihingen,[1] UAViD [49], and UDD5 [50]. Moreover, we validate our LDSeg in specific application scenarios, including building extraction (i.e., WHU^{Aerial} [51], Inria [52], and xBD [53]), road extraction (i.e., CHN6-CUG [54], DeepGlobe,[2] and SpaceNet [55]) and water extraction WBS-SI[3] dataset.

B. Experimental Settings

1) Metrics: To assess the performance of our proposed LDSeg, we utilize the mean intersection over union (mIoU) metric, which is a critical performance metric for semantic segmentation models. For single-class extraction tasks, the intersection over union (IoU) of the foreground class is adopted as the evaluation metric.

2) Implementation Details: In the experiments presented in this article, we employed a transformer-based pretrained CLIP model. Unless stated otherwise, ViT-Base [24] with a patch size of 16 is used as our visual backbone, and our code framework is built on mmsegmentation [58]. In the image processing pipeline, input images are resized to a long side of 448 pixels, followed by sliding window inference using a 224×224 pixel window and a stride of 112 pixels. Our HSE strategy employs GPT-3 [45] to generate subcategories for each target class, and some inappropriate subcategory names are manually removed. In the ablation study, we found that the SFR is insensitive to the number of features used for repairing singular points, as long as they are within an appropriate range. Therefore, considering simplicity and generalizability, in the SFR strategy, we used the same parameters in the experiments across all datasets.

3) Baseline: The adaptation of the original CLIP [9] for semantic segmentation, as introduced in Section III-A, results in poor performance. To highlight our contributions, MaskCLIP [21] is employed as the default baseline.

C. Performance Analysis

To rigorously evaluate the segmentation performance, we compare LDSeg against a set of state-of-the-art approaches on seven semantic segmentation benchmarks. As detailed in Table I, the comparison group includes representative remote sensing methods such as SegEarth-OV [40], as well as seven advanced training-free OVSS models originally developed for natural images: MaskCLIP [21], GEM [23], SCLIP [37], NACLIP [38], ClearCLIP [56], and ProxyCLIP [57]. Additionally, two recent fine-tuning-based approaches, OVRS [34] and TPOV-Seg [35], are incorporated to provide a holistic perspective.

[1] https://www.isprs.org/education/benchmarks/UrbanSemLab
[2] http://deepglobe.org
[3] https://www.kaggle.com/datasets/shirshmall/water-body segmentation-in-satellite-images

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TABLE I

EVALUATION RESULTS (MIOU, %) OF OUR METHOD AND OTHER METHODS. OUR RESULTS ARE MARKED IN GRAY. THE **BEST** AND SECOND BEST RESULTS ON EACH DATASET ARE HIGHLIGHTED

Method	Potsdam	LoveDA	iSAID	Vaihingen	UAVid	UDD5	DLRSD	Average
Training-free Methods:								
CLIP [9]	13.63	11.52	6.91	10.25	10.47	7.75	12.39	10.42
MaskCLIP [21]	26.41	25.58	14.16	24.69	27.78	29.36	18.21	23.74
GEM [23]	28.39	31.33	19.87	24.65	35.23	41.85	27.17	29.78
SCLIP [37]	33.35	30.40	19.31	27.64	38.31	44.10	28.22	31.62
NACLIP [38]	32.86	30.78	19.53	28.78	38.65	43.19	28.09	31.71
ClearCLIP [56]	32.85	29.84	18.82	27.30	37.51	42.02	27.93	30.91
ProxyCLIP [57]	35.24	33.38	21.44	30.64	42.55	47.45	29.26	34.28
LDSeg	43.08	35.00	22.39	37.51	42.65	46.97	32.74	37.19
Fine-tuning Methods:								
OVRS_{DLRSD} [34]	27.47	-	39.09	33.71	-	-	**85.98**	-
OVRS_{iSAID} [34]	19.94	-	**87.87**	20.82	-	-	20.67	-
TPOVRS_{LoveDA} [35]	30.10	**45.05**	-	27.55	-	-	-	-
Feature Upsampler:								
SegEarth-OV [40]	35.43	35.84	21.64	29.16	43.23	50.56	30.78	35.23
LDSeg-UP	**47.08**	38.88	21.54	**40.59**	**46.35**	**50.71**	35.30	**40.06**

1) General Semantic Segmentation: The quantitative results in Table I demonstrate the superiority of the proposed method across diverse scenarios. Among the training-free methods, LDSeg establishes a new state of the art, achieving an average mIoU of 37.19%. This performance significantly outperforms the vanilla CLIP baseline (10.42%) and surpasses the previous best-performing method, ProxyCLIP [57], by a margin of 2.91%. Notably, LDSeg exhibits robust generalization capabilities, ranking first on the challenging Potsdam and Vaihingen datasets while maintaining competitive results on others.

To showcase the robustness and scalability of our approach, we integrated LDSeg with SegEarth-OV, a dedicated remote sensing method, demonstrating its function as a training-free enhancement module. Specifically, we adopted the feature upsampler from SegEarth-OV [40] to mitigate issues from low-resolution features, such as shape distortion and boundary inaccuracies, without requiring extra training. The resulting variant, LDSeg-UP, achieved an average +2.87% mIoU improvement over the original LDSeg across multiple datasets, reaching a peak average mIoU of 40.06%. Furthermore, LDSeg-UP consistently outperformed the original SegEarth-OV [40], yielding a commanding average +4.83% mIoU advantage. This outcome robustly verifies the effectiveness, flexibility, and scalability of our training-free framework when enhancing existing remote sensing algorithms.

While fine-tuning-based methods (e.g., OVRS) can achieve high scores on their specific target datasets (e.g., 85.98% on DLRSD), their limited evaluation on nontarget domains confirms they fundamentally sacrifice cross-domain generalization. In contrast, LDSeg and LDSeg-UP deliver consistent, high-quality results across all evaluated domains without requiring any dataset-specific training or parameter fine-tuning, thus confirming their superior transferability and practical utility for genuine open-vocabulary segmentation tasks in remote sensing.

2) Specific Category Segmentation: Table II compares the mIoU values for segmenting specific categories (building, road, and water extraction) among the proposed LDSeg method and several others across multiple datasets. For building extraction, LDSeg outperforms the other methods on the WHU^{Aerial} [51], Inria [52], and xBD [53] datasets, achieving mIoU values of 49.24%, 43.91%, and 36.16%, respectively. For road extraction, the CHN6-CUG [54], DeepGlobe,[4] and SpaceNet [55] datasets have mIoU values of 34.89%, 20.23%, and 23.98%, respectively. In water extraction on the WBS-SI[5] dataset, LDSeg also achieves the best performance, with an mIoU of 51.6%. Notably, LDSeg outperforms the second-best method by a significant margin, with mIoU improvements of 8.69% on the WHU^{Aerial} [51] dataset, 4.8% on the DeepGlobe[6] dataset, and 3.23% on the WBS-SI[7] dataset. These results consolidate the effectiveness of LDSeg in category-specific segmentation, demonstrating its superiority over competitors across different datasets and tasks, with considerable potential for optical remote sensing semantic segmentation.

3) Qualitative Comparison: Qualitative results of MaskCLIP, ClearCLIP, SegEarth-OV, and LDSeg are presented in Fig. 7. Several key observations are summarized as follows. First, the prediction results of MaskCLIP contain substantial noise and exhibit significant prediction errors. A typical instance of such errors is the random distribution of mispredictions in flat road regions. Second, ClearCLIP

[4]http://deepglobe.org

[5]https://www.kaggle.com/datasets/shirshmall/water-body segmentation-in-satellite-images

[6]http://deepglobe.org

[7]https://www.kaggle.com/datasets/shirshmall/water-bodysegmentation-in-satellite-images

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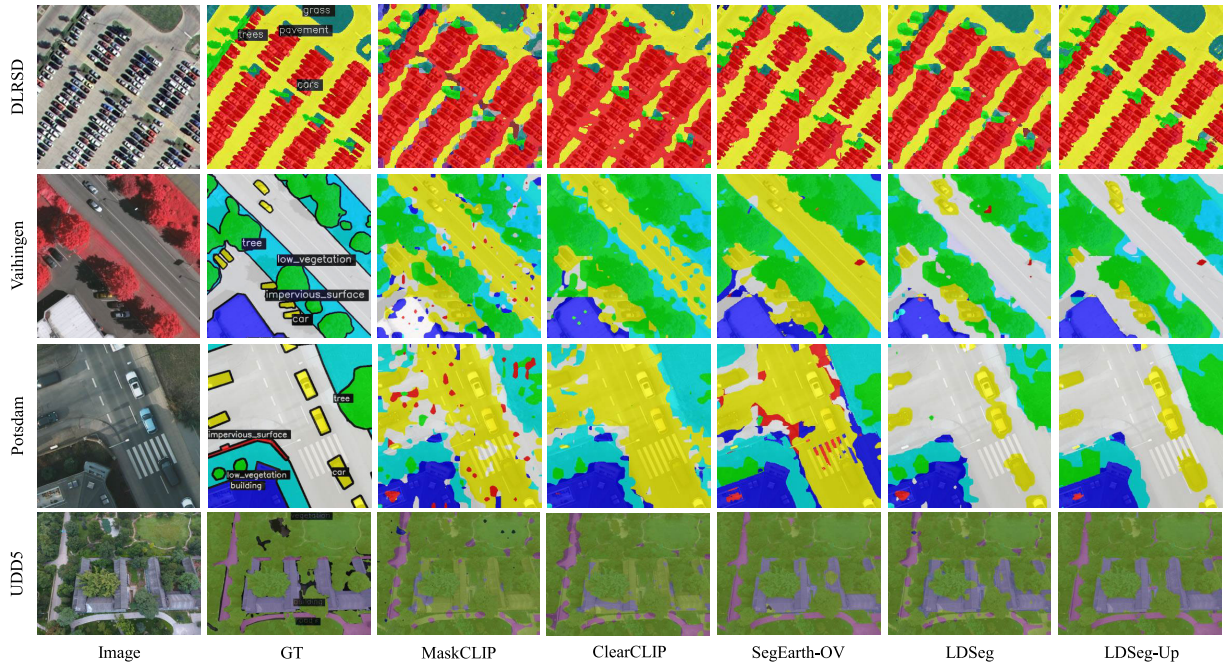


Fig. 7. Qualitative performance comparison across training-free OVSS methods, evaluated on DLRSD, Vaihingen, Potsdam, and UDD5 datasets. (Zoomed-in view digitally to better examine object edge and small objects.)

TABLE II

COMPARISON (MIOU, %) OF OUR LDSEG AND OTHER METHODS ON SPECIFIC CATEGORIES SEGMENTATION. OUR RESULTS ARE MARKED IN GRAY. THE BEST RESULTS ON EACH DATASET ARE **BOLDED**

Method	Building Extraction			Road Extraction			Water Extraction
	WHU^{Aerial}	Inria	xBD^{pre}	CHN6-CUG	DeepGlobe	SpaceNet	WBS-SI
CLIP [9]	16.32	18.32	16.04	5.25	3.37	6.55	18.25
MaskCLIP [21]	36.08	30.73	29.16	25.29	15.43	21.80	40.49
GEM [23]	24.99	29.03	20.78	15.19	4.92	12.41	41.21
SCLIP [37]	37.13	39.07	29.13	31.23	6.88	19.34	46.55
NACLIP [38]	40.55	41.15	30.32	32.95	8.04	21.05	48.37
ClearCLIP [56]	38.21	39.97	30.14	30.91	6.68	18.50	46.39
LDSeg	**49.24**	**43.91**	**36.16**	**34.89**	**20.23**	**23.98**	**51.60**

enhances the accuracy of category prediction but still lacks precise localization capability. Its predicted masks are discontinuous with poor boundary fitting, and an illustrative example of this limitation is the inferior connectivity of road regions in its results. Third, SegEarth-OV is capable of generating finer masks that fit the targets well. Nevertheless, it suffers from considerable prediction noise in its outputs. Fourthly, LDSeg can predict categories correctly and achieves higher recognition accuracy for ambiguous categories. Furthermore, LDSeg-UP, built on the basis of LDSeg, yields masks with improved fitting precision.

4) Performance Under ViTs With Different Scales: To evaluate the performance under ViTs with different scales, we conduct additional experiments on ViT-B/16 and ViT-L/14. The results are presented in Table III. Even with the ViT-L/14 configuration, our method continues to show improvements over other approaches.

5) Remote Sensing VLMs: As presented in Table IV, we evaluate the performance of various domain-specific CLIP models on multiple remote sensing datasets, including RemoteCLIP [11], GeoRSCLIP [12], and SkyCLIP [59]. These models share the same architecture as the original CLIP

TABLE III

COMPARISON (MIOU, %) OF OUR LDSEG AND OTHER METHODS UNDER DIFFERENT SIZE VITS. ^{bg} INDICATES THE EXCLUSION OF THE "BACKGROUND" CLASS. OUR RESULTS ARE MARKED IN GRAY. THE BEST RESULTS ON EACH DATASET ARE **BOLDED**

Method	ViTs	$Potsdam^{bg}$	$LoveDA^{bg}$	$iSAID^{bg}$	$UAVid^{bg}$
ViT-B/16	CLIP [9]	16.80	17.00	9.50	13.78
	SCLIP [37]	39.78	46.49	36.36	48.28
	NACLIP [38]	38.95	46.14	35.89	48.05
	ClearCLIP [56]	39.16	45.77	36.71	47.03
	LDSeg	**49.36**	**52.83**	**44.61**	**54.25**
ViT-L/14	CLIP [9]	10.41	14.48	4.96	8.59
	SCLIP [37]	34.61	47.46	48.18	46.10
	NACLIP [38]	36.24	47.69	47.93	46.15
	ClearCLIP [56]	35.08	47.19	48.59	45.38
	LDSeg	**55.21**	**51.47**	**53.94**	**53.66**

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TABLE IV

COMPARISON (MIOU, %) OF OUR BASELINE ACROSS MULTIPLE DATASETS WHEN ADOPTING VANILLA CLIP AND REMOTE SENSING CLIPS. ^{bg} INDICATES THE EXCLUSION OF THE "BACKGROUND" CLASS. THE BEST RESULTS ON EACH DATASET ARE **BOLDED**

Backbone	Train Data	CLIP Type	Potsdam^{bg}	LoveDA^{bg}	iSAID^{bg}	UAVid^{bg}
ViT-B/32	400M	CLIP [9]	38.97	38.81	30.94	38.43
	0.8M	RemoteCLIP [11]	23.22	**52.66**	45.47	23.08
	5M	GeoRSCLIP [12]	30.65	46.33	**49.46**	38.80
	2.6M	SkyCLIP [59]	**40.96**	46.14	36.30	**39.49**
ViT-L/14	400M	CLIP [9]	35.36	47.42	48.91	46.98
	0.8M	RemoteCLIP [11]	30.89	55.43	**63.39**	32.40
	5M	GeoRSCLIP [12]	34.21	**55.47**	53.71	**47.65**
	2.6M	SkyCLIP [59]	**36.48**	45.35	48.49	44.99

TABLE V

EFFICIENCY COMPARISON OF TRAINING-FREE METHODS

Method	Params(M) ↓	Speed(img/s) ↑	mIoU(%) ↑
CLIP	149.62	10.62	10.42
MaskCLIP	149.60	10.81	23.74
ProxyCLIP	235.42(+85.82M)	6.02	24.28
SegEarth-OV	149.92(+0.3M)	2.05	35.23
LDSeg	149.60	7.73	37.19

[9] but are trained with different volumes of remote sensing data (using RGB bands): 0.8 M samples for RemoteCLIP [11], 5 M for GeoRSCLIP [12], and 2.6 M for SkyCLIP [59].

As shown in Table IV, in most datasets, the models trained on RSI-text pairs generally perform better. This finding indicates that learning on domain-specific data effectively enhances the performance of the models. In addition, the comparison between CLIP and RemoteCLIP [11] highlights the importance of the data scale: while CLIP [9] is trained on massive 400 M samples, RemoteCLIP [11] leverages only 0.8 M samples, resulting in a noticeable performance gap.

6) Efficiency Comparison: We present the efficiency comparison on size 448×448 using a NVIDIA V100 GPU. As illustrated in Table V, LDSeg introduces no extra parameters in comparison with previous state-of-the-art ProxyCLIP [57] and SegEarth-OV [40], which introduce parameter overheads of 85.8 and 0.3 M, respectively. Regarding inference speed, LDSeg operates at 7.73 img/s, significantly outperforming ProxyCLIP [57] and achieving approximately **3.8**$\times$ the speed of SegEarth-OV [57] (2.05 img/s). Crucially, this efficiency does not compromise performance; LDSeg attains the highest mIoU of 37.19%. Thus, our method establishes the state-of-the-art tradeoff between computational cost and segmentation accuracy among training-free approaches.

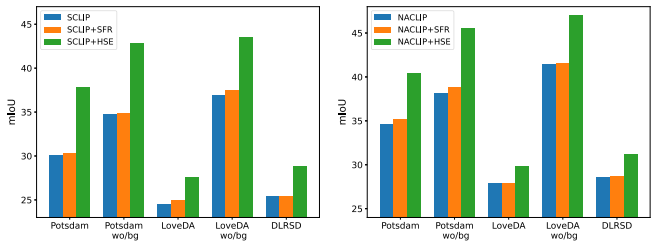
D. Ablation Study

1) Compatibility: In Fig. 8, we also report the performance when the latest advanced algorithms, SCLIP [37] and NACLIP [38], are combined with the SFR and HSE strategies proposed in this article. The findings demonstrate that our proposed methods are complementary to certain existing algorithms, offering additional performance gains, which substantiates the effectiveness and generalizability of our approaches.

TABLE VI

ABLATION RESULTS ON THE KEY COMPONENTS. THE BEST RESULTS ON EACH DATASET ARE **BOLDED**

	CSA	SFR	HSE	DLRSD	Potsdam^{bg}	LoveDA^{bg}	Vaihingen^{bg}	UAVid^{bg}
Baseline	×	×	×	18.21	32.19	38.70	28.91	37.60
+ SFR	×	✓	×	21.94	33.43	39.72	30.29	43.04
+ HSE	×	×	✓	23.82	42.12	44.10	36.80	42.69
+ CSA	✓	×	×	27.25	39.78	47.70	30.66	48.87
+ CSA + SFR	✓	✓	×	27.42	40.20	48.31	31.48	49.14
+ CSA + HSE	✓	×	✓	32.48	50.22	53.46	41.00	54.10
LDSeg	✓	✓	✓	**32.74**	**50.59**	**53.82**	**41.52**	**54.33**



(a) (b)

Fig. 8. Compatibility of our proposed method with state-of-the-art methods. (a) and (b) Performance gains of SCLIP [37] and NACLIP [38] when integrated with our proposed SFR and HSE, respectively.

2) Effect of Key Components: To validate the contribution of the key components, including CSA, SFR, and HSE, a systematic ablation analysis was conducted, with results detailed in Table VI. The baseline model, excluding these modules, achieved an average mIoU of 31.12%. Individual integration confirmed positive impacts: SFR improved mIoU to 33.68%, HSE reached 37.91%, and CSA provided the largest individual boost to 38.85%, underscoring the importance of our contributions. Analysis of combined modules showed strong synergy, notably CSA + HSE achieved 46.25%. The full LDSeg model, integrating all three, achieved the optimal overall performance with an average mIoU of 46.6%, securing the highest score across all five evaluated datasets (e.g., Potsdam^{bg}: 50.59% and UAVid^{bg}: 54.33%). This study conclusively validates the contribution and efficacy of the LDSeg framework's architectural designs.

In Fig. 8, we assess the performance of the advanced algorithms SCLIP [37] and NACLIP [38] after the incorporation of our SFR and HSE. The consistent improvements across datasets indicate that our methods are complementary to existing algorithms, validating the efficacy and generalizability of our techniques.

3) Tokens for Repair of Singularities: In Fig. 9, an ablation study is conducted on the number of tokens, denoted as N_f in (8), which are utilized to repair the features of singularities. The SFR method clearly exhibits robustness with respect to the number of tokens employed for generating the features of singularities.

4) Layers for the Detection of Similar Tokens to Singularities: We analyze the impact of attention maps from different shallow layers used to identify tokens similar to singular points on the set of datasets without a background class, including DLRSD, Potsdam^{bg}, LoveDA^{bg}, Vaihingen^{bg}, UAVid^{bg}, and

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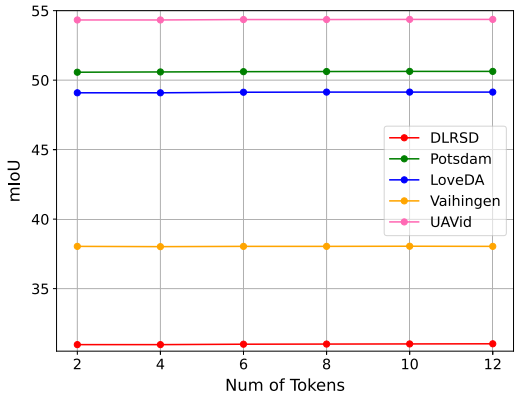


Fig. 9. Impacts of the number of tokens used to generate features of singularities in the SFR.

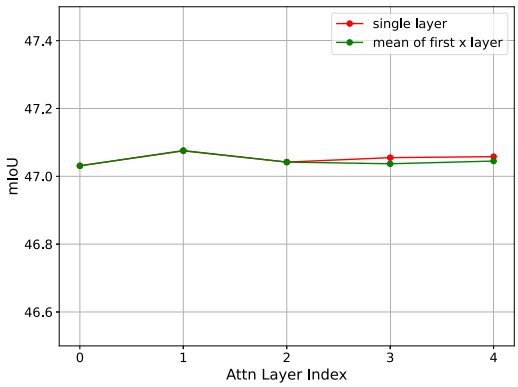


Fig. 10. Impacts of attention maps from different shallow layers employed to identify tokens similar to singular points.

UDD5[26], The mean results of all the datasets are illustrated in Fig. 10, which shows that the SFR method is robust with respect to shallow layers.

V. Limitations and Future Works

While LDseg demonstrates significant potential in the open-vocabulary segmentation of high-resolution optical imagery, three key limitations warrant discussion. First, its reliance on pretrained VLMs such as CLIP—designed for natural RGB imagery—introduces a semantic-text mismatch when handling domain-specific modalities (e.g., multispectral SWIR bands, SAR polarization features), as these lack textual metadata for alignment. Second, the absence of modality-specific encoders limits the preservation of unique spectral or SAR characteristics during fusion. Third, the scarcity of annotated text–modality pairs for non-RGB bands restricts the training of specialized remote sensing VLMs.

Future efforts will focus on three key directions: 1) designing modality-specific encoders (e.g., spectral CNNs and hybrid transformers) to extract domain-distinctive features while enabling cross-modal fusion; 2) developing adaptive alignment strategies (e.g., contrastive learning and attention mechanisms) to integrate complementary information from optical, multispectral, and SAR data; and 3) constructing domain-specific text-modality datasets via expert annotations or automated metadata generation to support VLM pretraining. Additionally, parameter-efficient fine-tuning methods (e.g., lightweight adapters) and standardized text-description protocols for spectral/SAR features will be explored to increase scalability and interoperability. These advancements aim to mitigate semantic-text mismatches, improve multimodal fusion interpretability, and extend VLMs to broader Earth observation applications.

VI. Conclusion

In this article, we explore the potential of foundational models for semantic segmentation in remote sensing imagery and introduce a training-free framework termed LDSeg for language-driven semantic segmentation of RSIs. Our LDSeg harnesses CLIP's ability to align visual and textual representations, coupled with its extensive fine-grained open-vocabulary capabilities, offering an attractive alternative to the limitations of traditional closed-set supervised training. LDSeg consistently achieved the best performance across all datasets used for performance evaluation. For the first time, we identify and thoroughly investigate the special intrinsic tokens within CLIP's visual representations that impede semantic segmentation. These special tokens are referred to as "singularities." A SFR strategy is introduced to rectify singularities, restoring their localization and semantic information. Furthermore, we innovatively investigate the impact of textual representation and introduce a straightforward HSE strategy that produces more precise text descriptions for images, fortifying the alignment of visual and textual representations. In conclusion, our LDSeg framework is highly practical because it requires no additional data or retraining, thus preserving the inherent generalizability of CLIP. Notably, our proposed SFR and HSE are orthogonal to existing approaches while functioning as plug-and-play modules to further enhance segmentation performance.

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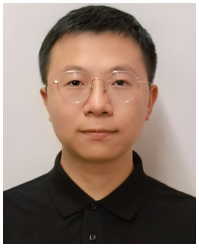
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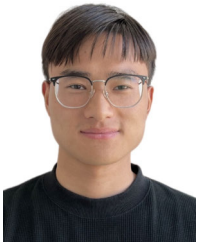
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